

■ B.4.2 Non-standard Argument Evaluation

There are a number of built-in *Mathematica* functions that evaluate their arguments in special ways. The control structure **While** is an example. The symbol **While** has the attribute **HoldAll**. As a result, the arguments of **While** are not evaluated as part of the standard evaluation process. Instead, the internal code for **While** evaluates the arguments in a special way. In the case of **While**, the code evaluates the arguments repeatedly, so as to implement a loop.

Control structures	arguments evaluated in a sequence determined by control flow (e.g. CompoundExpression , Repeat)
Conditionals	arguments evaluated only when they correspond to branches that are taken (e.g. If , Which)
Logical operations	arguments evaluated only when they are needed in determining the logical result (e.g. And , Or)
Iteration functions	first argument evaluated for each step in the iteration (e.g. Do , Sum , Plot)
Assignments	first argument only partially evaluated (e.g. Set , AddTo)
Pure functions	function body not evaluated (e.g. Function)
Holding functions	argument maintained in unevaluated form (e.g. Hold , Literal)

Built-in functions that evaluate their arguments in special ways.

Logical Operations

In an expression of the form $e_1 \&\& e_2 \&\& e_3$ the e_i are evaluated in order. As soon as any of the e_i are found to be **False**, evaluation is stopped, and the result **False** is returned. This means that you can use the e_i to represent different “branches” in a program, with a particular branch being evaluated only if certain conditions are met.

The **Or** function works much like **And**; it returns **True** as soon as it finds any argument that is **True**. **Xor**, on the other hand, always evaluates *all* its arguments.

Iteration Functions

An iteration function such as **Do**[f , { i , $imin$, $imax$ }] is evaluated as follows.

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- The iteration specification is evaluated. If it is not found to be of the form $\{i, \text{imin}, \text{imax}\}$, the evaluation stops.
- If the iteration variable i already has a value, a message is printed, and the evaluation stops.
- imin and imax are used to determine the sequence of values to be assigned to the iteration variable i .
- The iteration variable is successively set to each value, and f is evaluated in each case.
- The values assigned to i are cleared.

If there are several iteration variables, the same procedure is followed for each variable in turn.

Unless otherwise specified, f is not evaluated until a specific value has been assigned to i , and is then evaluated for each value of i chosen. You can use `Release[f]` to make f be evaluated immediately, rather than only after a specific value has been assigned to i .

Assignments

The left-hand sides of assignments are only partially evaluated.

- If the left-hand side is a symbol, no evaluation is performed.
- If the left-hand side is a function, the arguments of the function are evaluated, but the function itself is not evaluated.

The right-hand side is evaluated for immediate, but not for delayed, assignments.

Any subexpression of the form `Literal[expr]` that appears on the left-hand side of an assignment is not evaluated, but is replaced by the unevaluated form of expr before the assignment is done.